

WAEDENSWIL, Switzerland

WMO: 066730

Lat: 47.2208N

Lon: 8.6778E

Elev: 1596

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.2	(c) 21.5	(d) 10.1	(e) 9.8	(f) 19.8	(g) 14.6	(h) 12.2	(i) 23.5	(j) 21.6	(k) 42.8	(l) 18.7	(m) 42.1	(n) 6.3	(o) 30	(p) 0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.4	(c) 86.1	(d) 67.7	(e) 82.7	(f) 66.6	(g) 79.5	(h) 65.4	(i) 69.4	(j) 82.1	(k) 67.9	(l) 79.7	(m) 66.5	(n) 77.1	(o) 4.4	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	98.3	74.2	63.8	93.9	72.9	62.6	90.0	71.6	34.4	82.3	33.1	79.9	32.0	77.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.4	(b) 14.0	(c) 11.5	(e) 15.2	(f) 90.9	(g) 4.2	(h) 2.4	(i) 12.2	(j) 92.6	(k) 9.8	(l) 94.0	(m) 7.4	(n) 95.3	(o) 4.4	(p) 97.0		
(a) 17.4	(b) 14.0	(c) 11.5	(e) 13.6	(f) 72.3	(g) 4.2	(h) 1.5	(i) 10.6	(j) 73.4	(k) 8.1	(l) 74.3	(m) 5.8	(n) 75.1	(o) 2.8	(p) 76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	34.2	35.3	42.9	51.0	57.5	65.2	67.9	66.7	59.4	51.9	42.3	35.5		
	DBStd	13.80	7.32	7.45	7.42	7.12	6.80	6.81	5.88	5.92	5.80	6.29	6.30	6.41		
	HDD50	1975	492	411	236	73	13	1	0	0	3	55	242	449		
	HDD65	5492	956	830	684	419	248	81	38	51	184	406	682	913		
	CDD50	2307	1	1	17	105	244	458	555	517	285	114	10	0		
	CDD65	351	0	0	0	0	15	88	129	103	16	0	0	0		
	CDH74	2361	0	0	0	12	120	645	875	644	65	0	0	0		
	CDH80	613	0	0	0	1	15	173	247	175	2	0	0	0		
(14) Wind	WSAvg	4.0	4.2	4.3	4.4	4.3	4.4	4.1	3.9	3.7	3.5	3.3	3.7	4.1		
(15) Precipitation	PrecAvg	64.3	3.4	3.8	4.1	4.6	6.8	7.6	7.7	7.6	5.5	4.3	4.5	4.3		
	PrecMax	79.8	7.3	10.0	8.7	10.3	13.0	11.5	14.2	13.9	11.3	7.8	11.6	10.0		
	PrecMin	50.6	0.3	1.5	1.4	0.7	1.0	4.1	3.1	1.7	2.1	0.4	0.0	0.0		
	PrecStd	6.9	1.9	2.0	2.0	2.2	2.9	2.0	2.9	2.9	2.0	1.8	2.4	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	56.1	65.8	75.6	82.0	89.4	90.1	89.4	79.2	69.9	62.2	52.7		
		MCWB	48.8	46.3	51.0	56.8	62.9	69.0	68.9	68.4	65.4	59.8	52.6	46.7		
	2%	DB	50.1	51.5	61.3	71.0	77.5	84.8	86.1	84.8	75.9	65.9	56.7	48.9		
		MCWB	44.3	43.5	49.3	54.7	60.8	67.1	67.4	68.1	64.1	57.9	50.7	45.0		
	5%	DB	46.6	48.1	57.8	67.4	73.4	81.2	82.4	80.9	71.9	63.0	52.9	46.3		
		MCWB	42.6	41.9	47.7	53.2	58.9	65.8	66.6	67.0	62.3	56.7	48.9	43.3		
	10%	DB	43.5	44.9	54.1	63.3	69.6	77.4	78.9	77.0	68.5	60.2	50.5	44.1		
		MCWB	40.3	39.9	46.0	51.2	57.5	64.3	65.4	65.7	60.9	55.5	47.2	41.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	47.8	52.5	58.3	64.7	71.3	71.2	71.0	67.3	61.2	53.3	47.8		
		MCDB	55.2	54.7	62.2	72.5	78.0	85.9	86.1	83.7	76.7	66.5	58.4	51.5		
	2%	WB	45.3	44.7	50.3	55.9	62.5	68.7	69.1	69.2	65.0	59.3	51.2	45.5		
		MCDB	49.1	50.6	58.9	68.5	74.2	81.4	82.2	81.3	73.6	64.1	55.3	48.4		
	5%	WB	42.7	42.4	48.5	54.0	60.4	66.9	67.5	67.6	63.2	57.4	49.4	43.6		
		MCDB	46.1	47.2	56.2	65.3	70.0	78.3	79.3	78.7	70.5	61.8	53.1	46.3		
	10%	WB	40.5	40.2	46.6	52.2	58.5	65.3	66.3	66.2	61.5	55.9	47.5	41.6		
		MCDB	43.3	44.5	53.6	61.6	67.7	75.4	76.9	75.6	67.3	59.7	50.4	44.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	10.1	14.1	16.4	16.1	16.3	16.4	15.6	13.7	11.3	8.5	7.2		
		MCDBR	12.1	16.0	20.1	22.8	22.7	22.5	22.2	21.4	18.5	15.3	12.9	10.3		
	5% WB	MCWBR	8.5	10.0	11.1	11.2	10.5	9.4	9.1	9.2	8.9	8.3	7.3			
		MCWBR	11.6	14.7	18.5	21.2	20.0	20.4	19.9	19.7	17.0	13.7	12.3	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.309	0.348	0.388	0.404	0.411	0.413	0.404	0.375	0.345	0.311	0.280		
		taud	2.403	2.343	2.313	2.258	2.268	2.296	2.285	2.327	2.387	2.455	2.478	2.425		
	Ebn at Noon		257	271	276	274	273	271	269	266	264	256	244	244		
		Edn at Noon	22	29	34	40	41	40	40	37	32	26	20	19		
(44) All-Sky Solar Radiation	RadAvg		402	679	1063	1384	1566	1807	1761	1535	1162	741	425	332		
	RadStd		46	88	146	205	157	167	159	153	118	82	55	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A
(47) Regional (121 neighbors)		N/A	N/A	N/A	+1.26	+0.74	+0.59	N/A	N/A	+103	+29

Nomenclature: See separate page